

Section A: Inclusion / Exclusion Criteria									
Note to Programmer: Use the naming convention 'TimePeriod_Variable'									
Time Period	Variable	Label	Values	Definition	Code Lists	Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)	
INDEX	INDEX_DATE	Index Date	date	date of the first medical claim for multiple myeloma	Tab 42	At least one inpatient medical claim with a diagnosis code for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) OR ≥ 2 outpatient medical claims for multiple myeloma (any ICD-9-CM=203.0x or ICD-10-CM code=C90.0x) in any position on claim, on separate days within 30/60/90 days. (Suggest to program this as a macro, where 30, 60 & 90 day windows can be applied to the outpatient rule). The index date is the date of the first medical claim for multiple myeloma. Diagnosis codes for MM are in tab 42 "MM_dx code".		OK19012026	
INDEX	INDEX_YR	Index Year	date	year(index date)	Tab 42			OK19012026	
ID_PD	AGE_DX	Age year of index	numeric	year(index date) - birthyear	N/A			OK19012026	
BASELINE	CE_b	Baseline Continuous Enrollment	Binary	If six months of CE with medical and pharmacy benefits before the index date then =1; else =0.	N/A	Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled.		OK19012026	
FU_PD	CE_f	Follow-up Continuous Enrollment of at least 1 day	Binary	Only 1 day of CE is required in the follow up period then =1; else =0.	N/A	Patients with gaps in enrollment of ≤ 30 days are considered to be continuously enrolled		OK19012026	
FU_PD	CE_3most	Follow-up Continuous Enrollment of at least 3mos	Binary	Please flag patients if ≥ 3 months of CE starting on the index date and ending on the earliest of death, health plan disenrollment, or the study period end for patients who do not have an assigned date of death < 3 months after the index date then =1; else =0.	N/A	For patients with an assigned date of death < 3 months after the index date, the follow-up period ended at the assigned date of death. No gaps in enrollment are allowable. This patients may be subset on in the future.		OK19012026	
FU_PD	MM_FU_agents	Evidence of treatment with agents for the purpose of treating MM	Binary	Evidence of an FDA-approved MM oncology therapy in claims during the follow-up period =1; else =0. <i>Note : Receipt of corticosteroids only (eg, dexamethasone) will not be considered evidence of receiving an approved MM oncology therapy</i>	Tab 41	These will be identified with pharmacy claims (NDC) and medical claims (CPTs and HCPCS) codes		OK19012026	
BASELINE	MM_bl_agents	Evidence of MM oncology therapy in claims during the baseline period	Binary	≥ 1 medical or pharmacy claim for FDA-approved MM oncology therapy during the baseline period =1; else =0.	Tab 41	Evidence of any MM here is defined use ≥1 non-diagnostic medical claim for MM (ICD-9-CM=203.0x) or ICD-10-CM code=C90.0x) during the baseline period. This flag is used to help better identify smoldering patients. From Optum: "When we refer to a non-diagnostic multiple myeloma (or any disease) claim, what we are referring to is a claim where one of the multiple myeloma diagnosis codes is present, but there is at least one service line on the		OK19012026	

A	B	C	D	E	F	G	H	I
BASELINE	MM_baseline_diag	Evidence of any MM during the MM baseline period	Binary	Evidence of any MM (each of newly diagnosed and R/R MM), during the baseline period =1; else =0.	Tab 42 & 49	From Optum: "When we refer to a non-diagnostic multiple myeloma (or any disease) claim, what we are referring to is a claim where one of the multiple myeloma diagnosis codes is present, but there is at least one service line on the claim that doesn't equal a diagnostic code. The idea is that a physician/facility might order labs for testing/diagnostics for multiple myeloma and on the claim they have to put something for the diagnosis so they will put what they ordered the lab to test for. e.g., diagnosis code for multiple myeloma. If the resulting facility claim had a diagnosis code for multiple myeloma and only one service line on the claim and that service line indicated a laboratory test, then we can't be certain that this is a true multiple myeloma diagnosis because the claim is suggesting that only testing was done. Alternatively, if the claim had the diagnosis codes for multiple myeloma and two service lines: the first line for testing and the second line a HCPCS code for medication or a CPT code for physician care management then we will accept that claim as a "non-diagnostic claim" Indication: multiple myeloma " Patients with either ≥ 1 inpatient or ≥2 outpatient ICD-9-CM or ICD-10-CM codes on separate days, within 30 days, for the same primary tumor type and/or metastatic cancer. Only the first of the 2 codes is required to occur inside the baseline period.		OK19012026
BASELINE	MM_baseline_other	Evidence of another cancer in the baseline period	Binary	Evidence of another cancer in the baseline period =1; else =0.	Tab 44			OK19012026
FU_PD	MM_FU_other	Evidence of another cancer in the FU period	Binary	Evidence of another cancer in the follow up period =1; else =0.	Tab 44	Also flag patients with >= 2 outpatient claims on separate days within 60 or 90 days for consideration Patients with ≥ 2 ICD-9-CM or ICD-10-CM codes on separate days, within 30 days, for the same primary tumor type and/or metastatic cancer. ≥ 1 of medical claim with a diagnosis, procedure, or revenue code indicating pregnancy or childbirth during the patient identification period. (baseline or follow-up period)		OK19012026
ID_PD	Pregnant	Evidence of pregnancy	Binary	Evidence of pregnancy =1; else =0.	Tab 45	Evidence of clinical trial participation during the patient identification (study period) will be flagged		OK19012026
ID_PD	CT_part	Clinical trial participation	Binary	Clinical trial participation =1; else =0.	Tab 46	When date of death is only available at month-level granularity, the date will be generalized to the middle of the month or year, respectively (e.g., a date of death in June 2014 will be treated as 15 June 2014). For patients with only death year available, the death date is assumed to be July 15th year of the year they died. For patients with only death year available and their year of death is the same year as their index date, the death date is assumed to be July 15th of that year if the index date is on or before July 15th. The death date is assumed to be December 31st of		OK19012026
FU_PD	Death_dt	Death Date	Date	Date patient died.				OK19012026

A	B	C	D	E	F	G	H	I
FU_PD	Death_dt	Death Date	Date	Date patient died.		When date of death is only available at month-level granularity, the date will be generalized to the middle of the month or year, respectively (e.g., a date of death in June 2014 will be treated as 15 June 2014). For patients with only death year available, the death date is assumed to be July 15th year of the year they died. For patients with only death year available and their year of death is the same year as their index date, the death date is assumed to be July 15th of that year if the index date is on or before July 15th. The death date is assumed to be December 31st of that year if the index date is after July 15th. Note: When the flag CE_3mos=1 is applied health plan disenrollment or the study period for patients who do not have have an assigned date of death <3mos after the index date. When the flag CE_3mos=1 is applied, follow up end date will be earliest of death date, health plan disenrollment or the study period for patients who do not have have an assigned date of death <3mos after the index date.		OK19012026
FU_PD	ENDDATE	Follow Up End Date	Date	min(Death_dt, study period end date)				OK19012026
FU_PD	ENDDATE_CE	Follow Up End Date among CE pts	Date	min(Death_dt, date of health plan disenrollment, study period end date)		When the flag CE_3mos=1 is applied, follow up end date will be earliest of death date, health plan disenrollment or the study period for patients who do not have have an assigned date of death <3mos after the index date.		OK19012026
FU_PD	FU_DAYS	Length of follow up period (days)	numeric	The number of days between a patient's ENDDATE and their INDEX_DATE + 1				OK19012026
FU_PD	FU_DAYS_CE	Length of follow up period (days) among CE pts	numeric	The number of days between a patient's ENDDATE_CE and their INDEX_DATE + 1		Follow up time among patients who are continuously enrolled (without gaps in enrollment) in the FU period.		OK19012026

Section B: IE Criteria

Note to Programmer: Apply each criteria as a flag that can be carried forward or merged onto a later dataset. Helps make the cohort dynamic and flexible to change future study requirements.

Apply criteria in the order specified by the criteria number. Ensure that all flag are able to implemented independently rather than be executed sequentially/interdependent

Criteria No.	Criteria Type	Criteria Description	Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)
0	INCLUSION	≥ 1 medical claims for multiple myeloma in any position on claim	Serves as the base to the entire cohort. This flag will always be applied in the universe of the dataset.		OK19012026
1	INCLUSION	At least one inpatient medical claim with a diagnosis code for multiple myeloma (any ICD-9-CM code 203.00-203.99).	If criteria 1 flag is not implemented, none of the criteria will rely on the value of a patient's 1.		OK19012026
2	INCLUSION	Patients at least 18 years of age in the index year.			OK19012026
3	INCLUSION	Patients are required to have evidence of an FDA-approved anti-oncology therapy in claims.	Patients with gaps in enrollment of 3-30 days are considered to be continuously enrolled.		OK19012026
4	INCLUSION	Patients are required to have evidence of an FDA-approved anti-oncology therapy during the baseline period.	Patients with gaps in enrollment of 3-30 days are considered to be continuously enrolled.		OK19012026
5	INCLUSION	Flag patients with ≥ 2 months of continuous enrollment starting on the index date + follow up period.	Patients with ≥ 2 months of continuous enrollment starting on the index date + follow up period.		OK19012026
6	INCLUSION	Flag patients with ≥ 1 day of continuous enrollment starting on the index date + follow up period.	Patients with ≥ 1 day of continuous enrollment starting on the index date + follow up period.		OK19012026
7	EXCLUSION	Patients with ≥ 2 non-diagnostic diagnosis codes on separate days, within 30 days, for the same condition.	This is evidence of another cancer in the baseline period. Only the first of the 2 codes is used.		OK19012026
8	EXCLUSION	Patients with ≥ 2 non-diagnostic diagnosis codes on separate days, within 30 days, for the same condition.	This is evidence of another cancer in the baseline period. Only the first of the 2 codes is used.		OK19012026
9	EXCLUSION	Patients with ≥ 2 non-diagnostic diagnosis codes on separate days, within 30 days, for the same condition.	This is evidence of another cancer in the baseline period. Only the first of the 2 codes is used.		OK19012026

Section C: Final Study Population

Dataset	Label	Definition	Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)
ELIG_COH	A patient never dataset containing those patients who were not in the study at baseline.	The patient never dataset containing those patients who were not in the study at baseline.	Apply restrictions as a flag that can be carried forward or merged onto a later dataset.		OK19012026